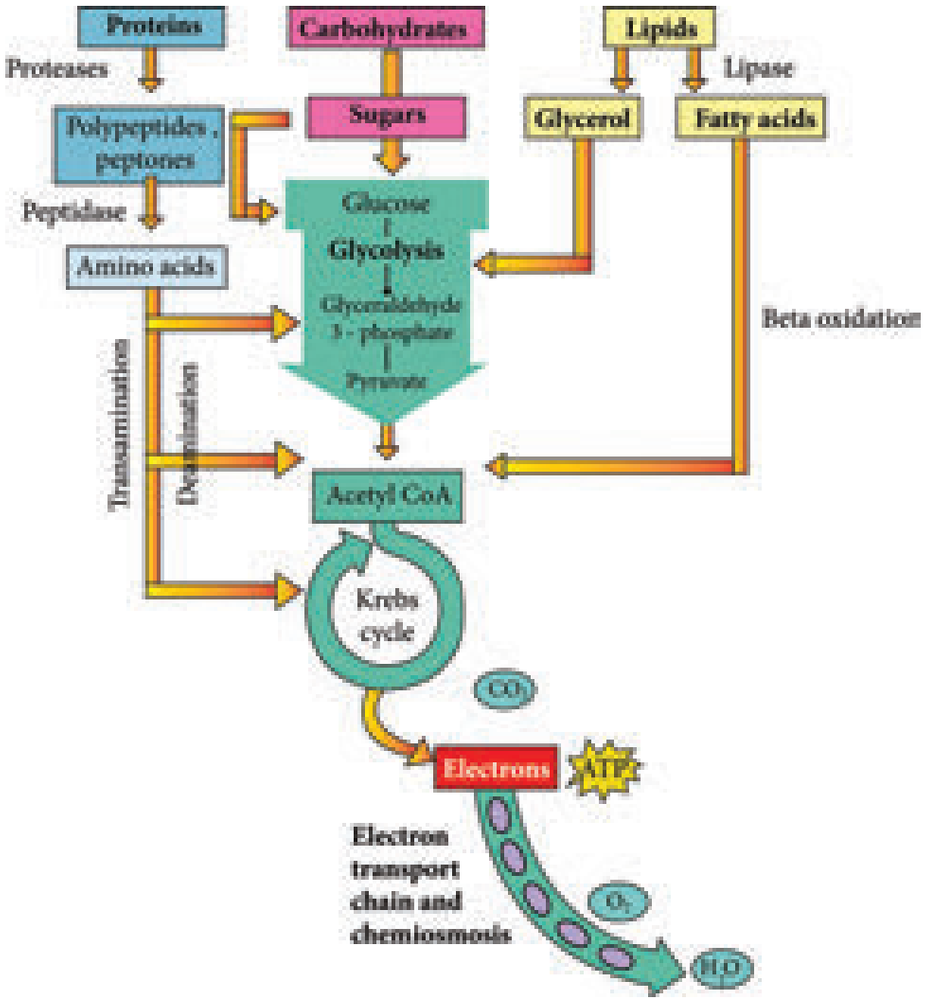


Figure 4.8: Overall Metabolism of Protein , Carbohydrates and Lipids

4.9 Fermentation

In 1856 fermentation, reaction was first demonstrated by Louis Pasteur in yeast. The study of fermentation and its practical uses is named as Zymology. Any energy releasing metabolic process that takes place only under anaerobic condition is called fermentation. It can also be defined is a metabolic process that release energy from a sugar or other organic molecule. It does not require oxygen or an electron transport system, and uses an organic molecule as the final electron acceptor. Fermentation reaction yields only a small amount of energy (2 ATP).

Anaerobes do not use an electron transport chain to oxidize NADH to NAD^+ and therefore use fermentation as alternative method to maintain a supply of NAD^+ for the proper function of normal metabolic pathways. Facultative anaerobes can use fermentation under anaerobic condition and carryout aerobic respiration when oxygen is present. Fermentation reoxidizes NADH to NAD^+ by converting pyruvic acid into various organic acids.

During fermentation, NADH is converted back into the coenzyme NAD^+ so that it can be used again for Glycolysis

TN_GOVT_XII_Micro_Biology_CH04.indd 58 04-03-2019 12:55:09